



MD-100 Windows Client

Module 01: Install the Windows client

© Copyright Microsoft Corporation. All rights reserved.



Module Agenda



Explore the Windows client



Examine Windows client editions and requirements



Evaluate Windows client installation methods



Upgrade and migrate Windows clients



Evaluate deployment methods

Lesson 1: Explore the Windows client



Lesson 1: Explore the Windows Client



Describe the Windows client



Explore what's new in Windows client



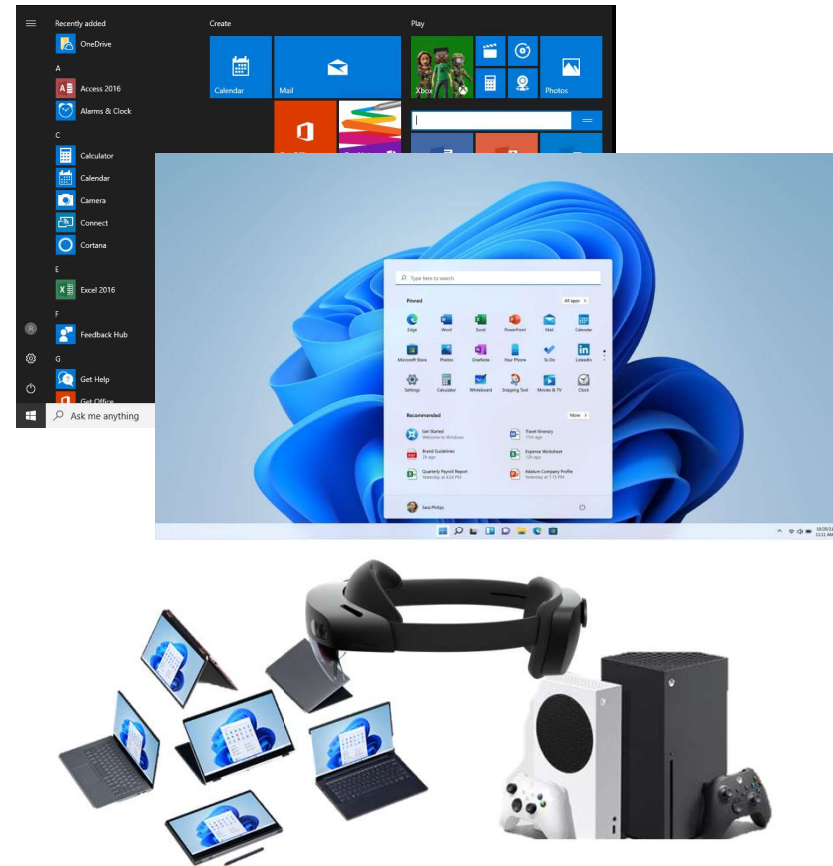
Evaluate the benefits of Windows client for small and medium-sized organizations



Explain desktop management in an enterprise environment

Describe the Windows client

- Multiple form factors and hardware
- Seamless transition for users of multiple devices
- Focus on cloud service integration
- Enhancements delivered through regular continuous updates
- Easier to implement updates and upgrades



Explore what's new in Windows client

Windows 10 improvements

- Windows as a Service
- Support for multiple device types
- Improved cloud integration
- MDM management and support for BYOD
- New and enhanced recovery tools and options
- Client Hyper-V and Windows Sandbox
- Broadband & VPN Enhancements
- Security enhancements

Windows 11 improvements

- New Start Screen
- Multiple UI Enhancements
- Snap Layouts and Snap Groups
- Widgets
- New Microsoft Store
- Support for Android Applications

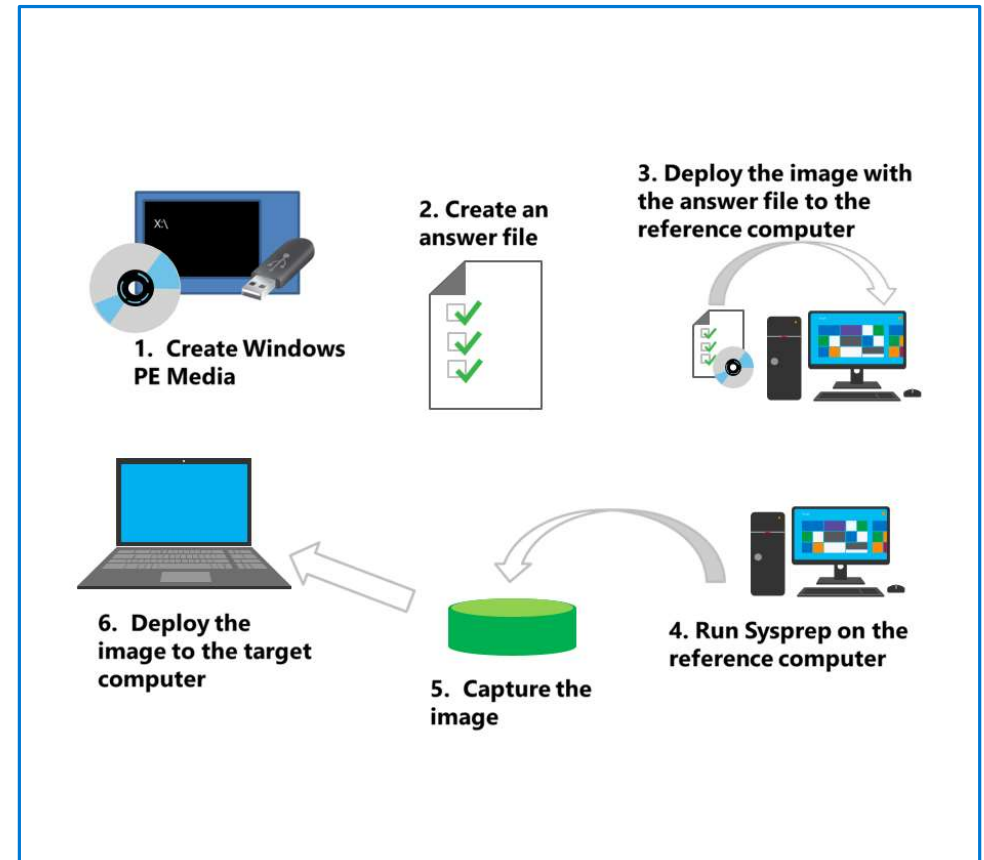
Evaluate the benefits of Windows client for small and medium-sized organizations

There are many reasons for small and medium-sized organizations to consider Windows 10 or 11, these include:

- Easier to use
- Continuous updates
- Improved device management
- Distribution of apps by using the Microsoft Store for Business
- More secure

Explain desktop management in an enterprise environment

- Simplified deployment of new and replacement computers.
 - Traditional Imaging
 - MDM Configuration Policies
- Consistent applications and hardware across the organization.
- Simplified inventory.
- Simpler software deployment.



Lesson 2: Examine Windows client editions and requirements



Lesson 2: Examine Windows client editions and requirements



Examine Windows client editions and capabilities



Select a Windows client edition



Examine Windows hardware requirements

Examine Windows client editions and capabilities

Edition	Consumer	Availability
Home	Individual/home use	Everybody
Pro	Small and medium-sized businesses, advanced users	Everybody
Enterprise E3 and E5	Large enterprises	Only available to Volume Licensing customers
Enterprise LTSC	Large enterprises	Only available to Volume Licensing customers
Pro Education	School staff, administrators, teachers, and students	Only available through academic Volume Licensing
Education	School staff, administrators, teachers, and students	Only available through academic Volume Licensing
IoT	Fixed purpose devices	Windows IoT Distributor

Select a Windows client edition

- **Form Factor**
 - Desktop/Laptop
 - Tablet/Hybrid
 - Xbox/HoloLens/SurfaceHub
- **Function**
 - End User
 - Kiosk/Shared
- **Management and infrastructure requirements**
- **Capabilities**
- **32/64-bit**

Examine Windows hardware requirements

Minimum recommended Windows 10 requirements	Minimum recommended Windows 11 requirements	Feature-specific requirements
• Processor: 1 GHz or faster processor or SOC • RAM: 2 GB (64-bit) • Hard disk space: 20 GB (64-bit) • Graphics card: DirectX 9 or newer with WDDM 1.0 driver • Display: 800x600	• Processor: 1 GHz or faster processor or SOC • RAM: 4 GB • Hard disk space: 64 GB • Graphics card: DirectX 12 or newer with WDDM 2.0 driver • Display: 720p, 9" or greater monitor, 8 bits per color • UEFI, Secure Boot capable • TPM 2.0	• Windows Hello: biometric features require compatible hardware • BitLocker requires a password, a TPM or a USB flash drive • Client Hyper-V: 64-bit system with SLAT capabilities and additional 2 GB of RAM • BitLocker to Go: USB flash drive • DirectStorage: NVMe SSD

Lesson 3: Evaluate Windows client installation methods



Lesson 3: Evaluate Windows client installation methods



Explore Windows client installation methods



Describe installation methods for Windows client



Examine the process of installing the Windows client



Evaluate new deployment options

Explore Windows client installation methods

Traditional methods

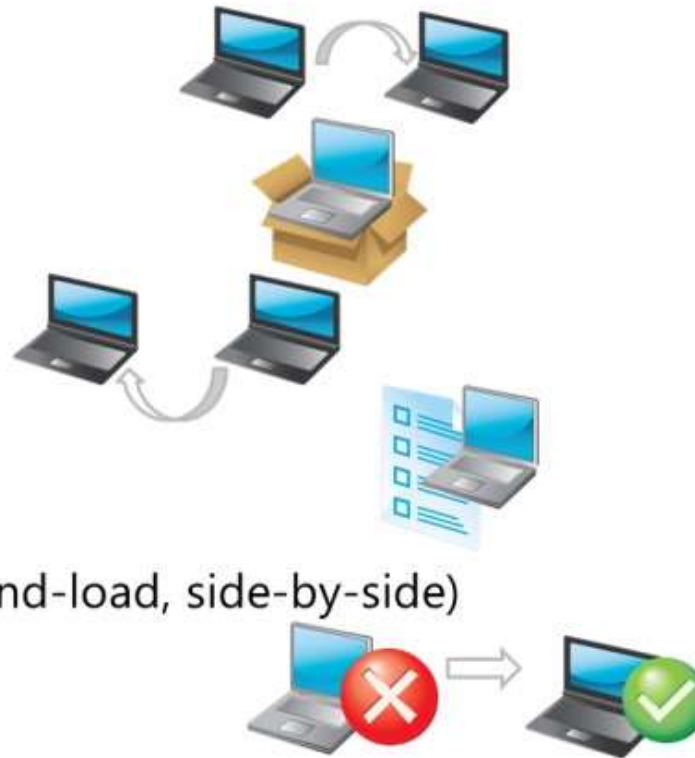
- Media (Setup.exe)
- Wipe and load

Modern Methods

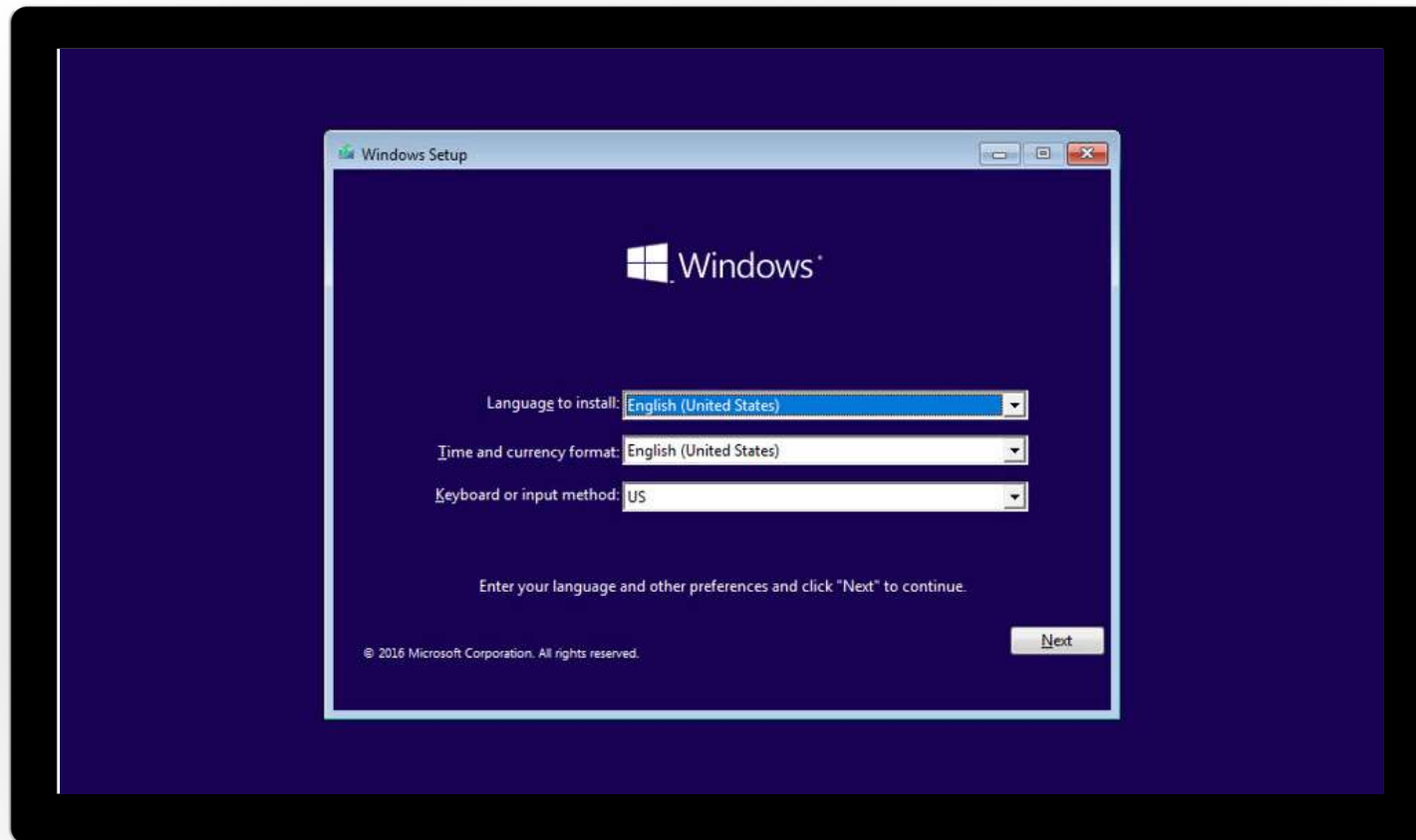
- In-place upgrade
- Re-configuration
- Provisioning
- Upgrade through Updates

Describe installation methods for Windows client

- In-place upgrade
- New deployment
- Refresh
- Provisioning
- Migration (wipe-and-load, side-by-side)



Examine the process of installing the Windows client

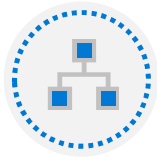


© Copyright Microsoft Corporation. All rights reserved.

Evaluate new deployment options



Deployment scenario



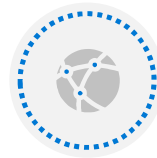
Centralized management



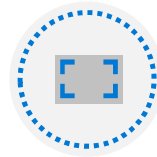
Required Drivers



Hardware and storage requirements



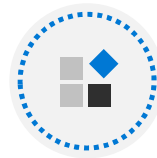
Network connectivity and capacity



OS features



Licensing and activation



Applications

Lesson 4:

Upgrade and migrate Windows clients



Lesson 4: Upgrade and migrate Windows clients



Examine the process of upgrading Windows clients



Explore supported upgrade paths



Examine the process of migrating Windows clients

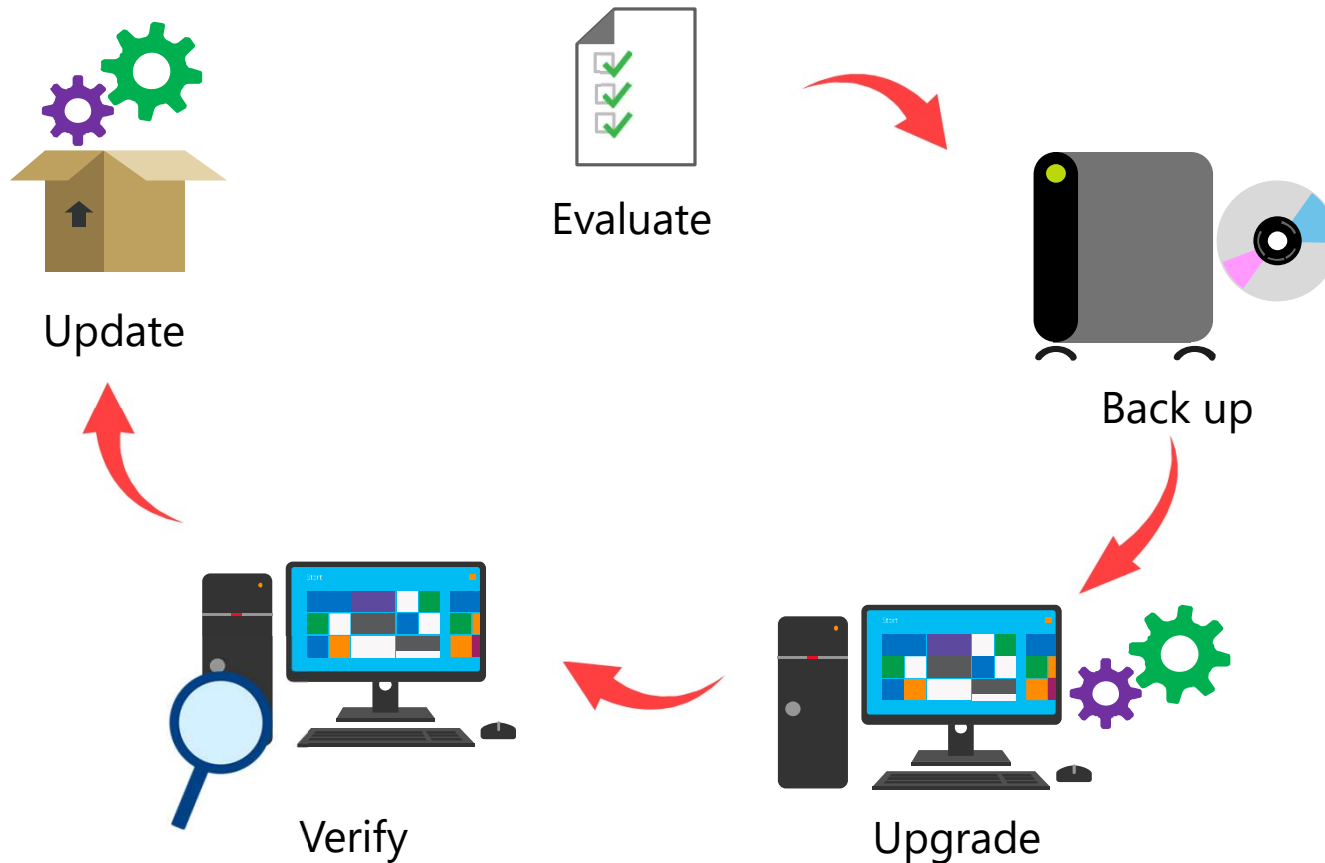


Compare upgrade and migration considerations



Evaluate common upgrade and migration scenarios

Examine the process of upgrading Windows clients



Explore supported upgrade paths

Initial OS	Windows 10/11 Home	Windows 10/11 Enterprise	Windows 10/11 Education
Windows 8.1	X		X
Windows 8.1 Pro/Pro Student	X	X	X
Windows 8.1 Enterprise		X	X
Windows 10 Home	X	X	X
Windows 10 Pro		X	X
Windows 10 Ent/Edu		X	X

Note: Some features might be removed when you upgrade to Windows 10/11

Explore supported upgrade paths

Does not require install for most scenarios

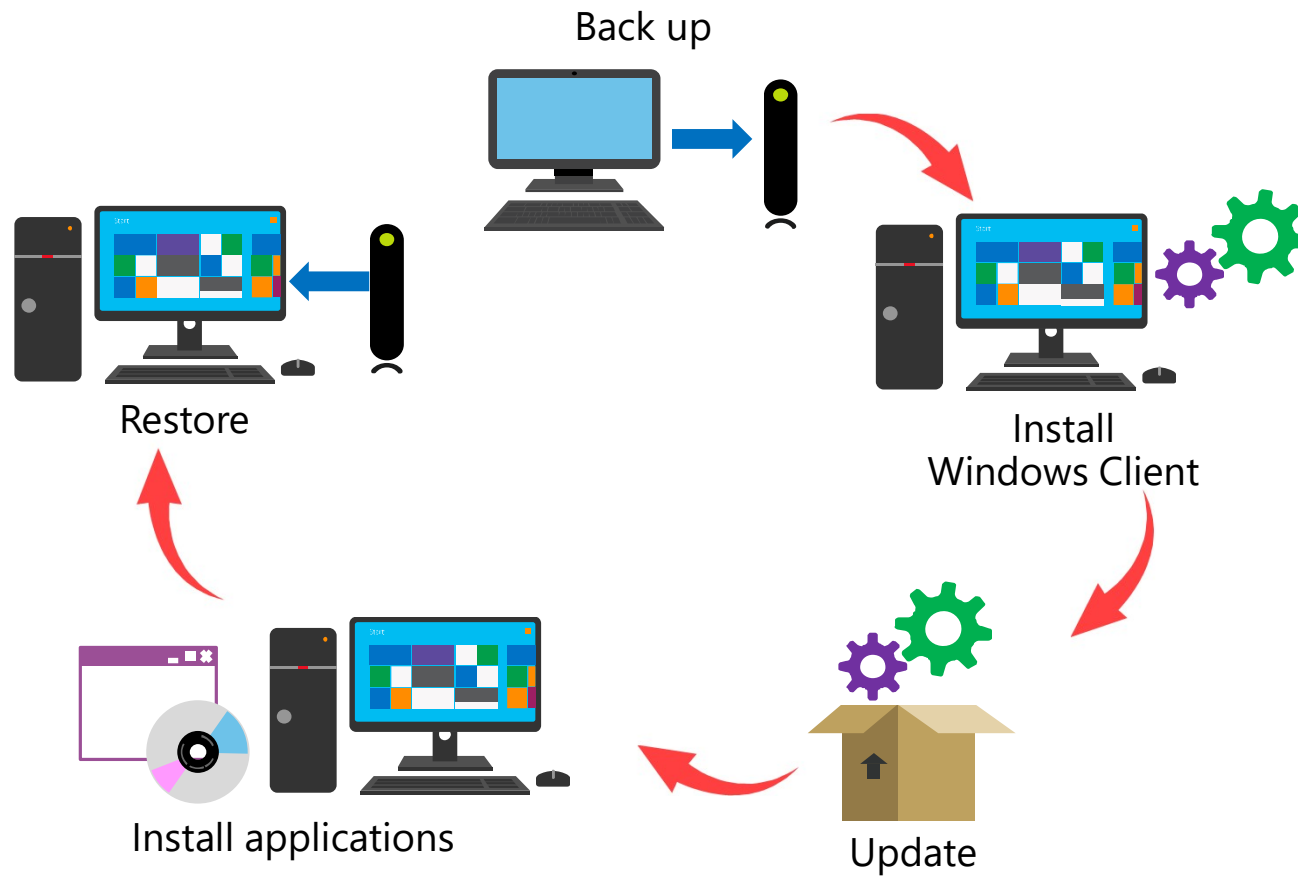
Path options depend on method

- Home -> Pro, Pro Workstation
 - Manual Key/Store Purchase
- Home -> Pro Edu, Edu
 - Manual Key/Store Purchase
 - Mobile Device Manager
 - Provisioning Package

- Pro/Pro Workstation/Enterprise -> Pro Workstation, Enterprise, Education
 - Mobile Device Manager
 - Provisioning Package
 - Microsoft Store for Business
- Upgrading may remove deprecated features

Note that upgrading the OS may remove deprecated features

Examine the process of migrating Windows clients



Compare upgrade and migration considerations



Advantages of in-place upgrades:

- Retains user settings, application settings, and files
- Preserves installed applications
- Does not require additional storage space for migration files
- Affects user productivity minimally, and preserves user settings and data
- Provides a simpler setup process
- Rollback is available in case of a problem



Advantages of migrations:

- Offers the opportunity to clean up existing computers and create more stable and secure desktop environments
- Allows for installation of any edition, regardless of the previous version that was running
- Provides the opportunity to reconfigure hardware-level settings before installation
- Malicious software does not migrate to the new installation



Disadvantages of in-place upgrades:

- Cannot start with standardized reference configurations
- Preserved applications may not work correctly after upgrading
- Remnant files or settings may contribute to performance and security issues
- Does not allow for edition changes
- Is only available on supported operating systems
- Computer must meet minimum hardware requirements



Disadvantages of migrations:

- Requires the use of migration tools, such as USMT
- Requires reinstallation of applications
- Requires storage space for user settings and files to be migrated
- May impact user productivity

Lesson 5: Evaluate deployment methods



Lesson 5: Evaluate deployment methods



Examine deployment options for Windows client



Explore imaging and autopilot



Explore deployment strategies



Explore deployment tools



Explore the User State Migration Tool



Explore client virtualization

Examine deployment options for Windows client

Modern

- Windows Autopilot
- In-place upgrade

Dynamic

- Subscription Activation
- Active Directory/MDM enrollment
- Provisioning Packages

Traditional

- Bare Metal
- Refresh
- Replace

Explore imaging and autopilot

- **Imaging.**
 - Creating a snapshot of a reference computer and deploying to multiple devices.
 - Traditionally used to support scaled deployments with consistent configurations.
 - Can be completely automated.
 - Images can be complex to create and manage.
- **Windows AutoPilot**
 - Reconfigures a Windows 10 or later client to an Out-of-Box-Experience (OOBE) while applying configurations and applications.

Explore deployment strategies



High-touch with retail media deployment. Insert the Windows client DVD and run the setup program for a manual deployment



Lite-touch deployment. Use MDT with Windows Deployment Services to automate most of the deployment, including applications, device drivers, and updates



Zero-touch deployment. Use MDT with Configuration Manager to deliver a fully automated deployment



Bring Your Own Device. Use MDM solutions such as Microsoft Intune to deploy and manage configurations and apps on user-owned devices.

Explore deployment tools

Tool	Function
DISM*	Capture, deploy, service, and manage Windows images
Windows SIM*	Create unattended installation answer files
Windows PE*	Minimal operating system used in Windows deployment.
Windows Configuration Designer*	Create provisioning packages

Tool	Function
USMT*	Migrate user settings
Volume Activation Management Tool (VAMT)*	Graphical tool used to automate and manage activation of Windows
Windows Performance Toolkit*	Produce in-depth performance profiles of OS and Apps.
Windows Assessment Toolkit*	Create jobs to perform tasks used in diagnosing issues
Windows Deployment Services	Enables the deployment of Windows operating systems
Microsoft Deployment Toolkit	For building and facilitating Windows OS deployments

*Included in Windows Assessment and Deployment Kit (ADK)

Explore the User State Migration Tool

Stores and migrates data and settings to another computer

- User Accounts
- User Files
- OS Settings
- Application Settings

Explore client virtualization

Client Hyper-V

- Client Hyper-V is a Windows Client feature that enables virtualization
 - Runs multiple operating systems on a single physical computer
- Client Hyper-V features:
 - Shares features with Hyper-V in Windows Server
 - Support for 32-bit and 64-bit VMs
- Client Hyper-V management tools:
 - Hyper-V Manager
 - Virtual Machine Connection tool
 - Hyper-V module for Windows PowerShell

Windows Sandbox

- Creates a Pristine Windows environment on the client.
- Does not require setup of Hyper-V or VHDs
- Temporary. All changes are lost when the Sandbox is closed.
- Must be enabled in Windows Features.

Practice labs

Deploying Windows using Windows ADK tools

Deploying Windows Desktop settings using a provisioning package

Provision a Kiosk using Configuration Designer

Migrating user state using USMT

Module review

